THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET B

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. $\sqrt[3]{3375}$ is: 1

(a) 15 **(b)** 16 **(c)** 14 **(d)** 17

2. $\sqrt{196}$ is equal to: 1

(a) 15 **(b)** 13 **(c)** 16 **(d)** 14

3. The number 0.0412 in standard (scientific) form is: 1

(a) 4.12×10^{2} **(b)** 4.12×10^{-3} **(c)** 4.12×10^{-2} **(d)** 41.2×10^{-2}

4. $(10^{3})^{5}$ equals: 1

(a) 30^{5} **(b)** 10^{15} **(c)** 10^{8} **(d)** 10^{243}

5. One billion is: 1

(a) 10^{12} **(b)** 10^{7} **(c)** 10^{6} **(d)** 10^{9}

6. The value of 1000 in base 4, in the decimal system, is: 1

(a) 63 **(b)** 64 **(c)** 65 **(d)** 66

7. 99 in Roman numerals is: 1

(a) XCVIII **(b)** C **(c)** CIX **(d)** XCIX

8. The Mesopotamian number system was based on: 1

(a) 5 **(b)** 60 **(c)** 10 **(d)** 20

9. Three angles of a quadrilateral are 65°, 85° and 110°. The fourth angle is: 1

(a) 110° **(b)** 80° **(c)** 100° **(d)** 90°

10. The diagonals of a parallelogram: 1

(a) bisect each other **(b)** are parallel

(c) are equal **(d)** are perpendicular

11. The smallest digit that replaces the blank in 81_4 so that the number is divisible by 3 is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 7

12. Is 6912 divisible by 9? 1

(a) Only if the last digit is even **(b)** Yes

(c) Only if the number is prime **(d)** No

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Write the sum of the first 9 odd natural numbers without adding. **2**

14. Simplify $(3^2)^4 \div 3^6$. **2**

15. Name two ancient civilisations that had their own number systems. **2**

16. List two properties of the diagonals of a rhombus. **2**

17. State the divisibility test for 11. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Find the cube root of 5832 by prime factorisation. **3**

19. Simplify: $\dfrac{3^7 \times 3^3}{3^4}$. **3**

20. Diagonals of a rhombus are 18 cm and 24 cm. Find the side and perimeter. **3**

21. Find the smallest number divisible by 10, 16 and 24. **3**

22. Apply Kaprekar's routine to 3524 until it reaches 6174. **3**

OR

Show that the number formed by writing any digit three times, such as 777, is divisible by 3 and 37. **3**

23. Find a Pythagorean triplet whose smallest member is 9. **3**

OR

Find the smallest number by which 180 must be multiplied to make it a perfect square. Also find the square root of the product. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Find the square root of $\dfrac{441}{784}$. **5**

(ii) Find $\sqrt[3]{\dfrac{343}{1728}}$.

OR

(i) Write the Pythagorean triplets obtained from $m = 5$, 7 and 9 using $\left(m, \frac{m^2-1}{2}, \frac{m^2+1}{2}\right)$. **5**

(ii) Verify one of them.

25. (i) Convert 111010_2 to decimal. **5**

(ii) Convert 58 to binary.

(iii) Add $1011_2 + 110_2$.

OR

Case study: Computers store data in binary. A byte has 8 binary digits. **5**

(i) How many different values can a byte store?

(ii) Write the largest value in decimal.

(iii) Convert 200 to binary.

END OF QUESTION PAPER — Total: 50 Marks

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